PERFORMANCE EVALUATION FOR ADAPTABLE MACHINE TRANSLATION METHOD

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Abstract

We will describe the Adaptable Machine Translation Method (ADMT) and the experiments used in the performance evaluation for ADMT, using translation examples of travel English. Our proposed method will need several adapted subsystems for the limited scenes. ADMT automatically produces the subsystems through the Machine Translation System which uses Inductive Learning with Genetic Algorithms (GA-ILMT) from sample data. ADMT decides the subsystem which corresponds to the source sentences by using Multi-Stage Analogy Decision Method (MSADM) in the analogy decision process. Then ADMT translates the source sentences using the selected subsystem. We performed the experiments using translation examples "on the plane", "at the airport" and "at the check-in". We confirmed that ADMT was adequate enough to obtain more correct and more qualitative translation results than two other commercial machine translation systems in theses three scenes of travel English.

1 Introduction

Computer networks are rapidly becoming popular. As a result, society is quickly becoming internationalized. However the language problem disturbs this internationalization process. To resolve this problem, the machine translation system is very effective. The main method of machine translation systems is the Rule-based method [1] . The Rule-based method cannot deal adequately with various linguistic phenomena due to its use of limited rules. Therefore its correct translation rate is still low and the quality is poor. Moreover the improvement of the correct translation rate of the Rule-based machine translation system is difficult when this system has many translation rules. It is difficult because, even though the developer of the system adds one translation rule to the dictionary for correcting an error, this rule may cause several new errors. To resolve this problem, the Example-based machine translation [2, 3, 4] has recently been proposed. The correct translation rate and the quality of the Example-based machine translation system can be high because this method adapts to data automatically. However, this method requires many translation examples in its development.

Therefore a machine translation system is not yet popular. The reason is that the context and background knowledge are too huge for a machine translation system to handle adequately. However the correct translation rate and the quality of machine translation system are high if the data which the system deal with, is limited[5]. The reason is that the system can adequately contain the context and background knowledge for the limited data. However the system which only has the context and background knowledge for the limited data, can not translate various kinds of data.

To resolve this problem, we proposed a method of Adaptable Machine Translation (ADMT) in this paper. Our proposed method needs several machine translation subsystems which are adapted in advance to several scenes respectively. First, the system decides the scene which the source sentences corresponds to, using the analogy between sample data and source sentences automatically. Then the system translate them using the machine translation subsystem adapted to its scene. Our proposal is that the correct translation rate and quality of ADMT are higher than the other machine translation systems because the data which our system's method deal with, is limited. Moreover, in ADMT, the system can deal with the various kinds of data because it can select the subsystem corresponded to the scene of the source sentences by using the analogy between the sample data and these source sentences.

In this paper, we propose ADMT and describe the experimental results of ADMT's performance evaluation, and offer some possible considerations. The Example-based machine translation system adapts to the source sentences slowly because this method requires many translation examples for its development. However ADMT adapts to the source sentences quickly because this method can select the subsystem corresponded to the scene of these source sentences. Moreover ADMT continues adapting to the following source sentences using its learning capability in detail.

2 Outline of ADMT

2.1 ADMT

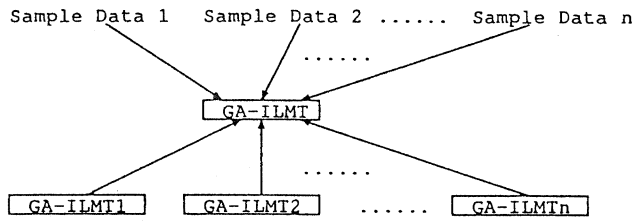


Figure 1: Method for Production of Machine Translation Subsystems

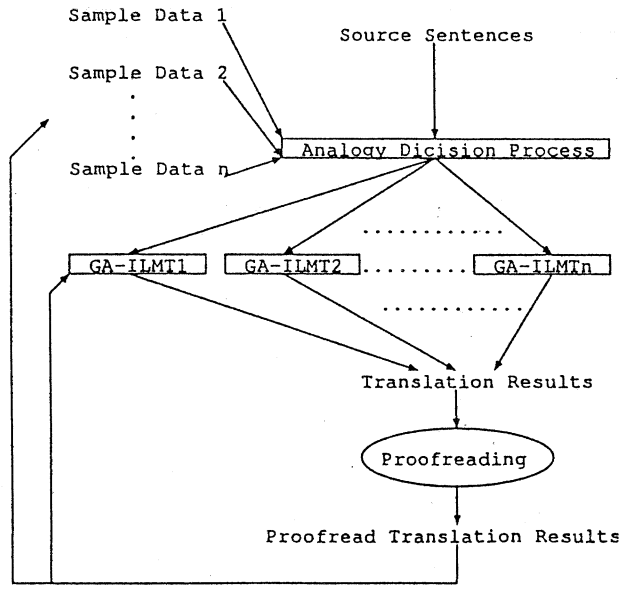


Figure 2: Outline of ADMT

Our proposed method requires machine translation subsystems adapted in advance to several scenes respectively. The subsystem are the system which has already adapted to a particular scene. Figure 1 shows the method of production of the adapted machine translation subsystems. In this figure, the sample data are translation examples in a particular scene. The translation example consists of a source sentence and a target sentence. We have already developed the method of Machine Translation using Inductive Learning with Genetic Algorithms (GA-ILMT)[6]. GA-ILMT can adapt to the particular scene using translation examples of the particular scene. Therefore ADMT produces automatically the subsystems using GA-ILMT from sample data. The subsystems are GA-ILMT1, GA-ILMT2, ..., GA-ILMTn.

Figure 2 shows the outline of ADMT. The correct translation rate and quality of ADMT are higher than the other machine translation systems for the limited data and ADMT can translate sentences of different scenes using various subsystems. ADMT decides which subsystem corresponds to the source sentences by using Multi-Stage Analogy Decision Method (MSADM) in the analogy decision process. Then ADMT translates them using the selected subsystem. The translation results are proofread by a user when they have some errors. GA-ILMT can acquire translation rules from translation examples. Namely GA-ILMT acquires translation rules from these proofread translation results. And they are added to the sample data which are used for the decision of the scene corresponding to the source sentences in the analogy decision process. Therefore ADMT keeps adapting to the source sentences.

2.2 GA-ILMT

Figure 3 shows the outline of GA-ILMT. First, a user inputs a source sentence in English. Second, in the translation process, the system produces several candidates of translation results using translation rules extracted in the learning process. Third, the user proofreads the translated sentences. Fourth, in the feedback process, the system determines the fitness value and performs the selection process of translation rules. In the learning process, new translation examples are automatically produced by crossover and mutation, and various translation rules are extracted from the translation examples by inductive learning. A translation example includes the source sentence and a translated sentence. There are two kinds of translation rules: those for sentences and those for words. The former are called sentence translation rules and the latter, word translation rules. Repetition of the above mentioned processes corresponds to generational replacement of the whole of system, and the system continuously evolves to higher quality translation.

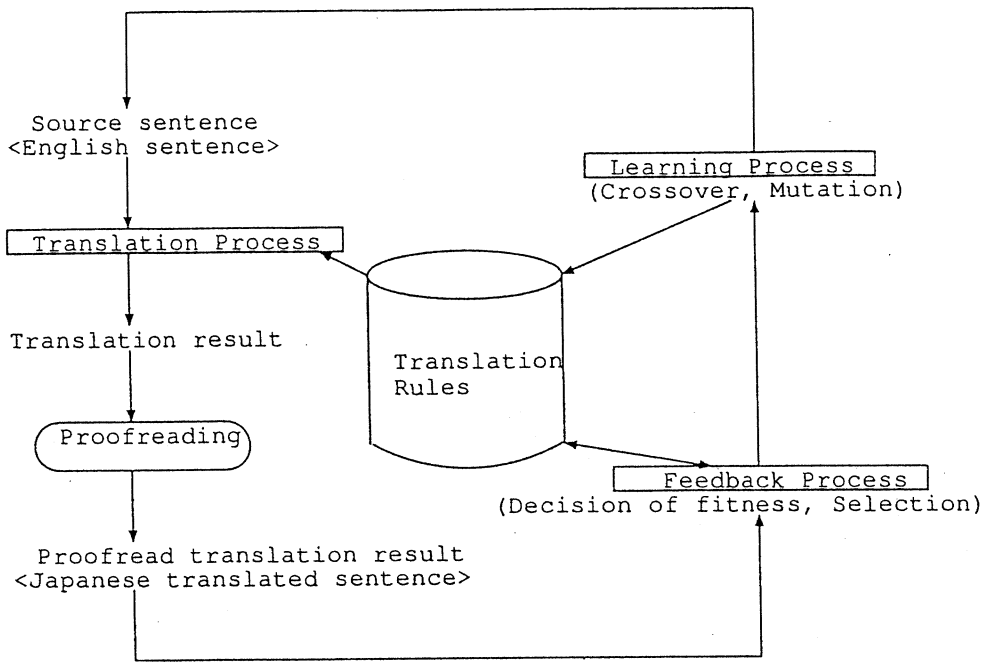


Figure 3: Outline of GA-ILMT

2.3 MSADM

MSADM decides the scene of source sentences using the analogy between sample data and source sentences. MSADM decides the corresponding words between a sentence of the sample data and a sentence of the source sentences in stages. The basic idea of MSADM is that the analogy between two sentences is high when there are many corresponding words. MSADM decides the corresponding words in order of the higher certainty of words. The procedure of MSADM is as follows:

1. the decision of words of the same position and same character string.
2. the decision of words of the same character string.
3. the decision of words of the same original form.
4. the decision of words of the same equivalent.
5. the decision of words of the same superclass concept.
6. the decision of words which both neighbors of them are the decided words.
7. the decision of words of the same part of speech.

Table 1 shows the dictionaries and tool used in each step.

The analogical degree is expressed by AD. AD is defined as follows:

$$\begin{aligned} AD &= (2.0 \times a1 + 1.8 \times a2 + 1.6 \times a3 + 1.0 \times a4 \\ &\quad +0.8 \times a5 + 0.4 \times a6 + 0.2 \times a7 \\ &\quad -0.5 \times a8) \times \alpha \end{aligned} \tag{1}$$

where

$$\alpha = \frac{100.0}{\text{Number of words of two sentences}}$$

the a1,a2,.....,a7 mean the number of pairs which the system decided on corresponding words between two sentences in each step respectively, and a8 means the number of pairs which the system could not decide.

Therefore AD between two identical sentences are 100.0.

3 Experiments

3.1 Data and Procedure

We evaluate the performance of ADMT using translation examples of three scenes of the English travel conversations from English to Japanese. These three scenes are "on the plain", "at the airport" and "at the check-in". We call "on the plane", "at the airport" and "at the check-in" Scene 1, Scene 2 and Scene 3 respectively.

Table 1: The Dictionaries and Tool Used by MSADM

Kind	Name	Number of words	Authors	Used step
Thesaurus	WordNet	126,516	Princeton University	3 5
Morphological dictionary	Oxford Advanced Learner's Dictionary of Current English	70,646	University of London	3
English-Japanese dictionary	*Eiwa-Waei Densaku Zisyo*†	55,000	*Shoji Kubo* (*Gizyutu Hyoron Sha*)	4
Equivalent list	ONE WORLD English Course 1 NEW EDITION (Word list)	389	*Kiyoshi Hasegawa* et al.	4
Equivalent list	NEW HORIZON English Course 1 (Word list)	397	*Akira Ota* et al.	4
Morphological analysis tool	Brill Tagger	1,429,865	Eric Brill	7

†Italic means Japanese.

Table 2: Numbers of Data

Number of books	Scene		
	1	2	3
1	111	103	75
2	83	96	83
3	58	155	48
4	78	160	60
5	78	75	120
6	41	69	66
7	44	39	41
8	68	164	44
9	13	84	50
10	39	94	23
Total	613	1039	510

The total number of data is 2162.

Table 3: Numbers of Data Used the Evaluation Experiment of MSADM

Kind	Number of books	Scene			Total
		1	2	3	
Sample	1 – 6	449	658	352	1459
Source Sentences	7	44	39	41	129
	8	68	164	44	276
	9	13	84	44	141
	10	39	94	23	156

In these experiments, a correct translation result defines an exact same character string or a very similar expression as that of a translator. These evaluation standards are very high. However translations like this are required by today's users.

The data for this experiment was taken from 10 English travel books for Japanese. These books are [7, 8, 9, 10, 11, 12, 13, 14, 15, 16]. We call these books No.1, No.2, ..., No.10 respectively. Table 2 shows the numbers of the data.

And we used two other machine translation systems which are developed by A and B companies for comparison with the system based on GA-ILMT. We call these two machine translation systems System A, System B respectively. The initial dictionary of GA-ILMT was empty each experiment. The conditions of the selection process were that the number of uses would be over 5 and the fitness value would be under 25%. The fitness value is defined as follows:

$$\text{Fitness value}(\%) = \frac{\text{The correct translation frequency}}{\text{The number of uses}} \times 100 \quad (2)$$

These thresholds were determined by a preliminary experiment. Translation examples were randomly changed by mutation, at a rate of 2%.

And we performed the evaluation experiment for the decision capability of a correct scene by MSADM. We used the data from No.1 to No.6 as the sample data, and the data from No.7 to No.10 as the source sentences in each scene.

And the system uses the upper four places of a word's concept number of WordNet in Table 1 in order to get the corresponding words which have the same superclasses in fourth step of MSADM. Table

3 shows the numbers of data used in the evaluation experiment of MSADM.

3.2 Results and Discussion

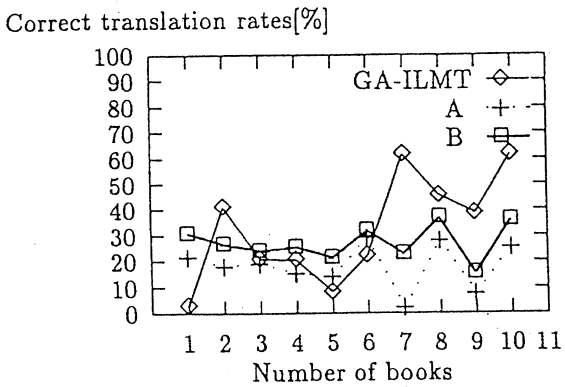


Figure 4: Changes of the Correct Translation Rate for the Data of Scene 1

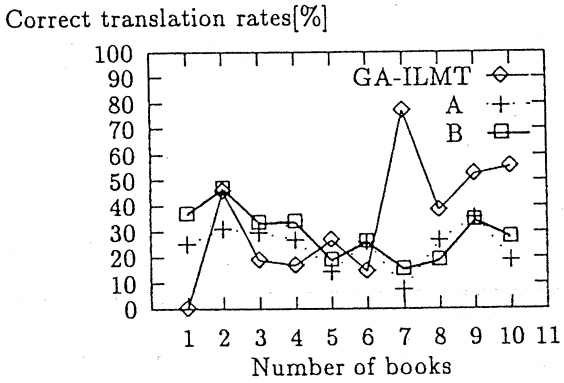


Figure 5: Changes of the Correct Translation Rate for the Data of Scene 2

Figure 4, 5 and 6 show the change of the correct translation rate for the data of Scene 1, Scene 2 and Scene 3 respectively. The total resulting data shows that the total correct rates of GA-ILMT, System A and System B are 20.9%, 24.0% and 29.5%. However the total correct translation rates after the seventh data are 47.4%, 22.8% and 24.6% respectively. The total correct rate of GA-ILMT after the seventh data is the highest among these three systems. Therefore we confirm that the learning capability of GA-ILMT is high enough so that this system can adapt to Scene 1, Scene 2 and Scene 3.

Moreover, these experimental results show that the machine translation system which can perform a contextual translation has been developed by GA-ILMT. For example, in Scene 1, GA-ILMT translated "Chicken, please" to "*Tori Ryouri ni shitekudasai.*"

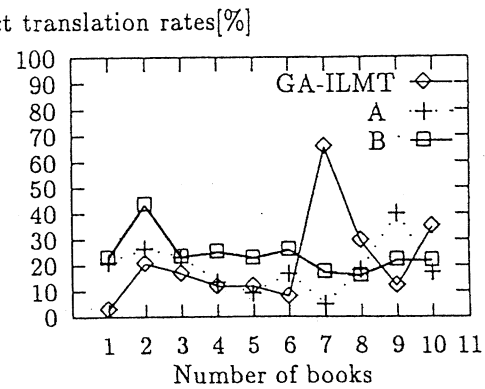


Figure 6: Changes of the Correct Translation Rate for the Data of Scene 3

(In Japanese "A dish of chicken, please.") System A translated it to "*Douka, Okubyoumono dearu.*" (In Japanese "Be coward, please.") and System B translated it to "*Chikin wo Onegai shimasu.*" (In Japanese "Chicken, please.") In the context of Scene 1, "chicken" means "a dish of chicken".

These results show that GA-ILMT is adaptable enough for these kinds of examples from English travel conversations. Therefore GA-ILMT records a higher-correct translation rate for English travel conversations than System A and System B after the seventh data. Furthermore the translation expressions of GA-ILMT are more natural than those of System A and System B. Therefore we confirmed that the correct translation rate and quality of GA-ILMT is adequately high after GA-ILMT adapted adequately to a particular scene.

Table 4 shows the results of the analogical degrees calculated by MSADM. Table 4 shows that all the AD's of the same scene between the sample data and the source sentences are the highest value among three scenes. These results show that MSADM decided the correct scenes for all the source sentences from No.7 to No.10. Therefore we confirmed that the correct translation rate and quality of ADMT was higher than System A and System B for the sentences of these three scenes in travel English.

4 Conclusion

We confirmed that ADMT was adequate enough to obtain more correct and more qualitative translation results than System A and System B in these three scenes in travel English. And we consider that ADMT can also be adequate enough to obtain similar performance for various scenes. Because, in these experiments, the initial dictionaries were empty of GA-ILMT and it could only learn the context from trans-

Table 4: Results of the Analogical Degrees Calculated by MSADM

Number of books	Scene of source sentences	Scene of sample data		
		1	2	3
7	1	**3923**	2527	2510
	2	2046	**3649**	2265
	3	1983	2337	**3745**
8	1	**5166**	2552	2193
	2	6864	**11555**	6708
	3	1936	2056	**3258**
9	1	**960**	744	647
	2	3523	**6554**	2858
	3	1283	1625	**2666**
10	1	**3156**	1321	1329
	2	3747	**6828**	3251
	3	979	1124	**1810**

lation examples through its own learning capability. And ADMT can decide automatically the scene for the source sentences using MSADM. However, we do not yet confirm the decision capability of the correct scenes for source sentences among several hundreds scenes by MSADM. In the future, we plan to experiment using the various data of many other fields to confirm it.

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